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Automated periapical lesion segmentation and area-based PAI indexing: a comparative deep learning study on periapical radiographs

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Abstract

Background Artificial intelligence can standardize periapical assessment, but few studies map pixel-level segmentation to clinically interpretable Periapical Index (PAI) categories under a unified protocol.

Methods We retrospectively assembled 900 anonymized periapical radiographs with expert masks and patient-level splits (train/val/test: 594/145/161). Four architectures (U-Net, ResUNet34, DeepLabV3, HRNet) were trained with identical preprocessing, augmentation, and binary cross-entropy (BCE) + Dice loss. On the independent test set, segmentation and image-level detection (sensitivity, specificity, precision, F1, AUC) were computed at a prespecified operating point. An area-based PAI score (aPAI) was derived from lesion-to-image area ratios using prespecified thresholds, providing a quantitative, size-based proxy for conventional PAI categories that does not incorporate qualitative radiographic features such as border definition or trabecular changes.

Results DeepLabV3 achieved the most balanced detection (accuracy 90.1%, sensitivity 92.8%, F1 91.8%), while HRNet yielded the highest specificity (87.5%) and precision (91.4%). Friedman/Wilcoxon analyses showed significant overall between-model differences. After Bonferroni correction (adjusted $\alpha = 0.0083$), only the DeepLabV3 versus U-Net comparison remained statistically significant for both binary lesion detection and aPAI classification (both $p = 0.0012$); no other pairwise differences reached the corrected threshold. aPAI classification accuracy ranged from 72.7% (U-Net) to 84.5% (DeepLabV3).

Conclusions Segmentation-based, area-derived PAI scoring is feasible and consistent across architectures. DeepLabV3 is preferable for screening workflows requiring high sensitivity, whereas HRNet minimizes false positives for confirmatory use. The unified pipeline provides an interpretable bridge from pixel probabilities to standard PAI categories.

Keywords Artificial intelligence, Deep learning, Periapical index, Periapical lesion, Radiograph segmentation

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Introduction

Artificial intelligence (AI) applications are increasingly integrated into dentistry, offering significant contributions to clinical diagnosis and decision-support processes [1]. In endodontics, AI technologies are being applied to reduce clinician workload and improve lesion detection accuracy in tasks such as radiographic image analysis, root canal anatomy characterization, and periapical pathology detection [2].

Deep learning (DL) methodologies, particularly convolutional neural networks (CNNs), form the foundation of recent advances in medical image analysis. CNN-based algorithms have demonstrated expert-level performance in disease detection, lesion segmentation, and image classification [3, 4]. In dentistry, CNNs have been successfully applied from caries and periodontal bone loss detection to periapical lesion assessment [5–7], enabling objective and reproducible analyses in dental radiographs [8, 9].

Within endodontics, periapical lesions represent one of the most common pathologies, characterized by bone destruction at the root apex due to pulpal infection [10]. Epidemiological studies indicate that apical periodontitis affects nearly half of the adult population worldwide [10]. Because these lesions often remain asymptomatic in early stages, they are typically detected incidentally during routine radiographic examinations [10, 11]. This underscores the clinical value of AI-assisted diagnostic systems, which may support practitioners in recognizing subtle periapical changes and facilitate more consistent diagnostic decisions [12].

The Periapical Index (PAI) remains the most widely used standardized tool for radiographic assessment of periapical status, categorizing lesions on a 5-point scale ranging from 1, indicating healthy periapical structures, to 5, reflecting severe periodontitis with acute features. Its validity is based on calibration against reference images of histologically confirmed cases, which has supported its adoption in both research and clinical settings for decades [13]. The PAI system not only enables consistent communication among clinicians but also provides a framework for longitudinal monitoring of periapical healing and disease progression.

Despite this clinical relevance, integration of AI with PAI-based assessment protocols has been limited. Preliminary attempts at automated PAI scoring exist, yet these are often restricted by small datasets, single-center designs, or narrow methodological scope, leaving important questions unanswered regarding which approaches are best suited for reliable, clinically interpretable scoring in routine practice.

In this context, two gaps remain insufficiently addressed in the literature. First, most AI studies [3, 14] on periapical radiographs have focused on lesion

detection or coarse localization, rather than on converting segmentation outputs into clinically interpretable severity categories. Second, few studies [7, 14] have compared multiple DL architectures under a unified tooth-level workflow with identical preprocessing, splitting, training, and evaluation conditions. Therefore, the present study aimed to develop an automated, tooth-level pipeline for periapical lesion segmentation on periapical radiographs and to derive an area-based Periapical Index (aPAI) from the segmented lesion area. The proposed aPAI was designed as a quantitative, reproducible, size-based proxy for conventional PAI scoring rather than a direct replacement, because it does not incorporate qualitative radiographic features such as trabecular change or lesion border characteristics. Four DL architectures -U-Net, ResUNet34, DeepLabV3, and HRNet- were compared under a single standardized protocol. Our null hypothesis was that these architectures would show comparable performance in both lesion segmentation and aPAI classification, with differences too small to be clinically meaningful.

Materials and methods

This was a retrospective diagnostic accuracy study using archived periapical radiographs. This study was conducted in accordance with the Declaration of Helsinki and approved by the Clinical Research Ethics Committee of Akdeniz University Faculty of Medicine (approval No. TBAEK-624). Informed consent was waived by the ethics committee due to the retrospective, anonymized design of the study. All patients provided general consent at the time of their clinical visits, acknowledging that their anonymized clinical data and radiographic images could be used for scientific research purposes. This study was reported in accordance with the STARD-AI (Standards for Reporting of Diagnostic Accuracy Studies-Artificial Intelligence) guidance [15]. The overall AI workflow and study design are summarized in Fig. 1 as a workflow overview.

Acquisition of periapical radiographic data

Periapical radiographic images were retrospectively collected from the archives of Akdeniz University Faculty of Dentistry between June 2015 and December 2024. All images were acquired using a KaVo Focus digital intraoral X-ray unit (KaVo Dental, Biberach, Germany) under standardized exposure conditions (70 kVp, 7 mA, 0.10 s). Size 2 photostimulable phosphor plates (PSP; 31 × 41 mm; Digora, Soredex Corporation, Tuusula, Finland) were used as receptors. Following exposure, latent images were scanned using a Digora Optime UV phosphor plate scanner (Soredex) and transferred to a secure workstation. All radiographs were obtained using the parallel technique with appropriate film holders to minimize

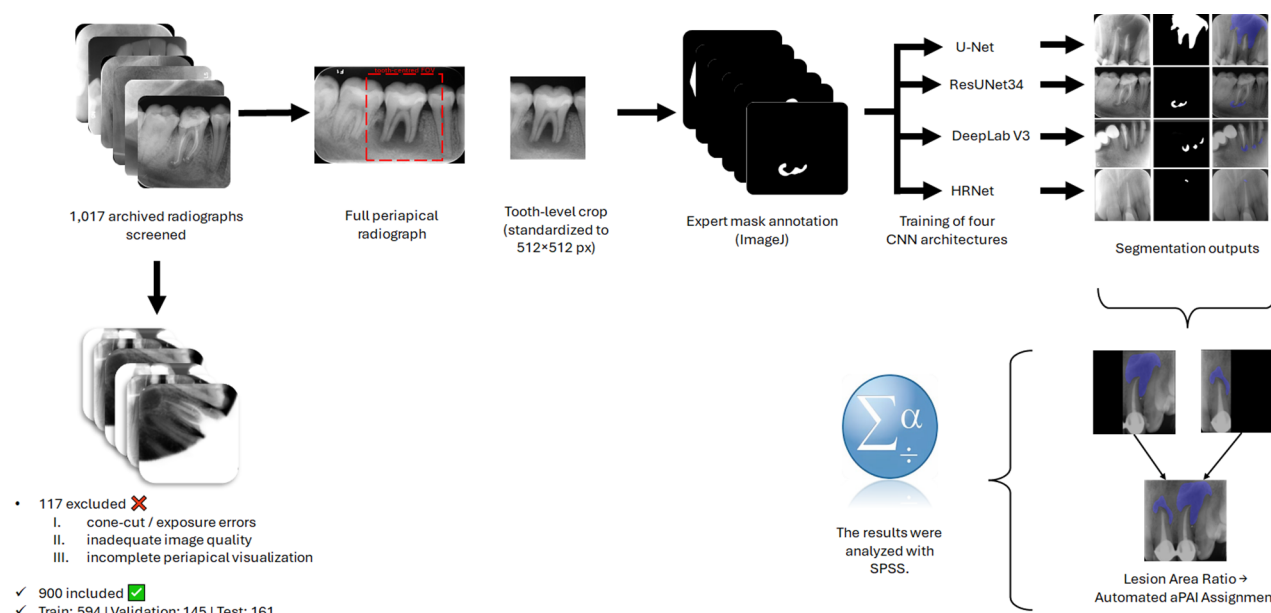


Fig. 1 Overview of the tooth-level AI workflow. Archived periapical radiographs were screened for eligibility, and diagnostically acceptable images were converted into tooth-centered crops that served as model inputs. Expert reference masks were created in ImageJ, and the dataset was divided at the patient level into training, validation, and independent test sets. Four DL segmentation architectures (U-Net, ResUNet34, DeepLabV3, and HRNet) were then trained to generate predicted lesion masks. These segmentation outputs were subsequently used to calculate lesion-to-image area ratios and derive the aPAI

geometric distortion and to ensure reproducible projection geometry. This standardized acquisition protocol reduces magnification variability and provides a more accurate representation of periapical lesion dimensions compared with the bisecting-angle technique.

All radiographs were initially saved in 8-bit grayscale format at a native resolution of 640×520 pixels. For DL analyses, the original archived radiographs were not used as full-image inputs. Instead, each model input consisted of a tooth-centered field-of-view that included the tooth of interest and its associated periapical region. These tooth-level crops were manually defined at native resolution to preserve periapical detail and then standardized to 512×512 pixels when required. Accordingly, the analytic unit in the AI pipeline was the tooth-level crop rather than the full radiographic image. In radiographs containing more than one tooth, separate crops were created only when distinct and spatially separable periapical lesions were present; these crops were then analyzed independently. Therefore, the aPAI score represents a per-tooth assessment rather than a regional aggregation across multiple teeth. No additional modifications were applied prior to annotation to avoid biasing the training process. Overlay and montage images were generated solely for visualization.

A total of 1017 archived periapical radiographs were screened for eligibility on an image basis rather than according to symptoms or referral indication. Radiographs of permanent teeth with diagnostically sufficient

quality and clear visualization of the periapical region were considered eligible for inclusion. After anonymization and quality review, 117 images were excluded because of severe motion artefacts, cone-cutting, saturation, inadequate visualization of the periapical region, or the presence of lesions with a different pathological basis, such as known cystic or tumoral conditions based on clinical and radiographic records. The final dataset therefore comprised 900 anonymized periapical radiographs, which were subsequently divided at the patient level into training ($n = 594$), validation ($n = 145$), and independent test ($n = 161$) sets to prevent information leakage.

Sample size calculation

Sample size was determined based on prior literature [7, 8, 14] reporting diagnostic accuracies ranging from 75 to 90% for periapical lesion detection. A minimum of 140 test samples was estimated to provide adequate power (> 80%) to detect clinically meaningful differences ($\geq 10\%$) in model performance at $\alpha = 0.05$. Our final test set ($n = 161$) with 97 positive cases and 64 negative cases exceeded this requirement, providing sufficient power for both primary analyses (accuracy comparisons using non-parametric tests). The sample size was also adequate for multi-class aPAI comparisons, with approximately 25 samples per category (range: 22–64).

Dataset partitioning and class distribution

The dataset was randomly divided into training ($n=594$), validation ($n=145$), and independent test ($n=161$) subsets using a reproducible Python random seed. To prevent data leakage, partitioning was performed at the patient level so that no patient contributed images to more than one subset.

Each subset contained a mix of healthy (PAI 1) and diseased (PAI 2–5) cases to maintain balanced class representation. The approximate distribution of PAI categories across subsets is shown in Table 1. The binary classification revealed 60% diseased cases and 40% healthy cases, which closely approximates general population prevalence estimates (~50%) and provides a more balanced dataset compared to typical clinical referral patterns. Although PAI 2–5 categories were equally distributed (approximately 15% each), healthy cases (PAI 1) were intentionally oversampled (40%) to improve model specificity for detecting true negatives. This balanced design enhances the clinical applicability of the models across diverse patient populations.

Ground-truth annotation

Lesion masks were manually delineated by 2 endodontists (each with >5 years of clinical experience) using ImageJ (NIH, Bethesda, MD, USA). For each image:

- The lesion region was outlined with the freehand tool according to consensus diagnostic criteria.
- Binary masks (512 × 512 px) were saved with identical filenames to enable one-to-one mapping with radiographs.
- Foreground pixels representing lesions were coded as “1,” background as “0.”

For the independent test set ($n = 161$), two expert endodontists -blinded to model outputs-assigned PAI scores according to Ørstavik et al. [13] and delineated lesion masks. Before establishing the consensus reference, inter-rater agreement was quantified as follows: for ordinal PAI, quadratic-weighted Cohen’s κ with 95% confidence intervals (and percent agreement); for segmentation

masks, the inter-rater Dice similarity coefficient with 95% confidence intervals. Disagreements were resolved by joint review to obtain the consensus ground-truth. For binary analyses, Positive = PAI 2–5 and Negative = PAI 1.

DL architectures

In this study, four widely used DL architectures were implemented and compared for the task of periapical lesion segmentation: U-Net, ResUNet34, DeepLabV3, and High-Resolution Network (HRNet). All models were trained and evaluated using identical data splits, augmentation strategies, loss functions, and evaluation pipelines to ensure fair and standardized comparison.

1. U-Net [16]

U-Net employs a classic encoder–decoder structure, comprising a downsampling (encoder) pathway and a mirroring upsampling (decoder) pathway, interconnected by skip connections at every corresponding resolution. Within each encoder block, 2 successive 3 × 3 convolutional layers are applied, each followed by batch normalization and ReLU activation, after which spatial reduction is accomplished using max pooling. The decoder path reverses this process through transposed convolutional layers to progressively increase spatial resolution. At each decoding stage, feature maps from the matching encoder block are concatenated to reintroduce fine-grained spatial information. The network produces the segmentation mask using a final 1 × 1 convolution, and this output is resized to 512 × 512 pixels through bilinear interpolation. Across the entire model, bias terms are omitted from all convolutional operations, and batch normalization is systematically employed after each convolution to enhance training stability.

2. ResUNet34 [17]

ResUNet34 integrates the U-Net architecture with a ResNet-34 backbone as the encoder, initialized with ImageNet-pretrained weights. Grayscale images were replicated to 3 channels to leverage ImageNet-pretrained backbones; ImageNet normalization was applied. The encoder extracts hierarchical features, which are progressively upsampled in the decoder using transposed convolutions and concatenated with corresponding encoder features via skip connections. An optional attention gating mechanism (attention block) is included in each decoder block to enhance the network’s focus on relevant lesion regions while suppressing background noise. The final mask is obtained by a 1 × 1 convolution and upsampling to 512 × 512 pixels. This design leverages the strong feature extraction capability of residual networks and

Table 1 Dataset distribution across PAI classification scores and data splits. PAI: Periapical Index according to Ørstavik et al. (13) classification system. PAI 1 represents healthy periapical tissues, while PAI 2–5 indicate increasing severity of periapical pathology

PAI Class	Train (n)	Validation (n)	Test (n)	Total (n)	%
PAI 1 (Healthy)	238	58	64	360	40.0
PAI 2	92	23	25	140	15.6
PAI 3	92	23	25	140	15.6
PAI 4	79	19	22	120	13.3
PAI 5	93	22	25	140	15.6
Total	594	145	161	900	100

benefits from both multi-scale context and focused lesion localization.

3. DeepLabV3 [18]

DeepLabV3 utilizes a ResNet-50 backbone and the Atrous Spatial Pyramid Pooling (ASPP) module. Grayscale radiographs were replicated to 3 channels to leverage ImageNet-pretrained weights, and ImageNet normalization was applied. The ASPP module processes the feature map with parallel atrous convolutions at different rates, enabling the detection of lesions of varying size. The final segmentation head outputs a single-channel probability map, which is upsampled to the original resolution for loss computation and evaluation. This architecture is particularly suitable for segmenting objects with variable size and shape, as is common in periapical lesions.

4. High-Resolution Network (HRNet) [19]

HRNet maintains high-resolution representations throughout the network by processing features in parallel across multiple resolutions and frequently exchanging information between them. The architecture consists of multiple stages with residual blocks, and at each stage, parallel branches operate at different resolutions, fusing features to enable robust multi-scale learning. A Convolutional Block Attention Module (CBAM) is integrated at the network's output stage to improve feature representation by applying both channel-wise and spatial attention mechanisms, thereby facilitating more precise localization of lesions. The segmentation head applies 2 sequential 1×1 convolutions, followed by upsampling to the input image size. HRNet's design allows for precise boundary delineation and robust detection of both small and diffuse lesions.

Standardized training and evaluation

All models were trained and validated under an identical protocol to ensure fair comparison. The same data splits (patient-level), preprocessing, augmentation strategies, loss functions, optimizer, and evaluation metrics were used across all architectures; only the network architecture differed.

Training Protocol: Inputs were resized to 512×512 pixels. A composite BCE+Dice loss was optimized using Adam optimizer (initial learning rate $1e-4$, batch size 4). Learning rate was adaptively reduced on validation plateaus using ReduceLROnPlateau scheduler. Early stopping was applied based on validation loss score (patience=15 epochs) to prevent overfitting. Training was performed on a Linux workstation with NVIDIA RTX 3060 GPU (12 GB VRAM) using mixed precision for computational efficiency. The checkpoint with

the highest validation Dice score was selected for final evaluation.

Implementation: PyTorch 1.9+, Albumentations 1.1+, and scikit-learn 0.24+ were used for model development, augmentation, and evaluation respectively.

Evaluation Metrics: On the independent test set ($n=161$), segmentation performance was assessed using Dice similarity coefficient and Intersection over Union (IoU). For image-level lesion detection, a prespecified operating point was applied: probability maps were binarized at 0.5 threshold with a minimum area filter of 131 pixels ($\sim 0.05\%$ of a 512×512 image) to classify images as "positive." At this operating point, sensitivity, specificity, precision, F1-score, and confusion matrices were computed. ROC curves and AUC were generated using a continuous decision score defined as the expected lesion area ratio (mean per-pixel probability across the image).

Complete implementation details, augmentation parameters are provided in Supplementary Methods.

It should be noted that image-level detection performance in this study was derived from the segmentation output rather than from a direct image-level classifier. Specifically, the binary lesion presence/absence decision was obtained by: thresholding the per-pixel probability map at 0.5; applying a minimum area criterion of 131 pixels ($\sim 0.05\%$ of image area) to suppress spurious detections; and classifying images as "lesion present" if the thresholded mask exceeded this area threshold. This segmentation-derived detection approach is distinct from end-to-end image classification methods and introduces dependencies on both probability threshold and area threshold selection.

Derivation of aPAI

The conventional PAI system described by Ørstavik et al. [13] is an expert-assigned ordinal score based on multiple qualitative radiographic features in addition to lesion size, including margin definition (diffuse vs. well-defined), radiolucency patterns, and trabecular bone changes. In contrast, aPAI is intentionally simplified and uses only the lesion-to-image area ratio as the classification criterion. This design enables automated scoring and reduces observer-dependent variability, but it cannot capture qualitative characteristics that may carry independent diagnostic information. Accordingly, aPAI should be interpreted as a standardized, size-based proxy for conventional PAI rather than a direct one-to-one equivalent.

For each predicted lesion mask, an aPAI was automatically assigned based on the proportion of the segmented lesion area relative to the total image area. This approach provides a quantitative, reproducible proxy for the conventional PAI scoring system, though it should be noted that traditional PAI assessment incorporates qualitative

radiographic features (lesion boundary characteristics, radiolucency patterns, and trabecular bone changes) that cannot be fully captured by area measurements alone.

Threshold determination

The classification thresholds (Table 2) were prespecified based on the theoretical relationship between PAI categories and expected lesion dimensions, rather than derived from the training data, to avoid information leakage. Thresholds were calculated assuming a standardized image size of 512×512 pixels (262,144 total pixels), with lesion area boundaries corresponding to approximate clinical descriptions of PAI categories: PAI 1 (no lesion, $<0.1\%$), PAI 2 (small periapical radiolucency, $0.1\text{--}0.5\%$), PAI 3 (moderate lesion with mineral loss, $0.5\text{--}3\%$), PAI 4 (well-defined radiolucency, $3\text{--}7\%$), and PAI 5 (severe radiolucency with exacerbating features, $>7\%$). These thresholds were applied identically across all model outputs using custom Python scripts, and the resulting aPAI scores were saved in .txt and .xlsx formats.

Manual PAI scoring and statistical analysis

For independent reference assessment, manual PAI scoring was performed by the same 2 independent, blinded endodontists who created the training masks, assigning scores to each of the 161 periapical radiographs in the test set according to classic radiographic criteria. For each test image, the lesion area (in pixels) and the corresponding PAI score were recorded by both observers. The predicted PAI scores obtained from the DL models were compared with these manual reference scores on a case-by-case basis. Correct and incorrect predictions, as well as lesion area agreements, were systematically documented.

Statistical analyses were performed using SPSS version 27.0 (IBM Corp., Armonk, NY, USA). Because all four models were evaluated on the same independent test cases, model comparisons were based on paired observations. As normal distribution of paired performance differences could not be assumed, non-parametric procedures were preferred. Overall, between-model differences were assessed using the Friedman test, followed by Wilcoxon signed-rank tests for pairwise comparisons with

Bonferroni-adjusted significance thresholds. Agreement between expert-assigned PAI scores and model-derived aPAI scores was evaluated using quadratic-weighted Cohen's kappa. In line with previous AI-based diagnostic accuracy studies in endodontic imaging [8, 11], binary detection performance was summarized using accuracy, sensitivity, specificity, precision, F1-score, and AUC; 95% confidence intervals were estimated using non-parametric bootstrap resampling (1,000 patient-level iterations). No samples were missing or excluded after data partitioning, and no indeterminate index-test or reference-standard results occurred in the test set.

Results

Four deep learning architectures (U-Net, ResUNet34, DeepLabV3, and HRNet) were evaluated on an independent test set for image-level periapical lesion detection and for deriving an aPAI from the predicted segmentation masks. The primary outcomes were binary classification performance (healthy: PAI 1 vs. diseased: PAI 2–5) and agreement in ordinal aPAI categories, alongside segmentation agreement metrics. Group-wise comparisons across models were performed using the Friedman test, followed by Wilcoxon signed-rank tests for pairwise comparisons with Bonferroni-adjusted significance thresholds ($\alpha = 0.0083$ for 6 pairwise comparisons in both binary lesion detection and aPAI classification). Representative qualitative examples (original radiographs, predicted masks, and overlays) are provided in Figs. 2 and 3. No samples were missing, lost, or excluded after test-set allocation, and no indeterminate results occurred for either the index tests or the reference standard.

Inter-rater agreement on the independent test set ($n = 161$) was quantified using Cohen's kappa (quadratic weights for ordinal PAI). For binary image-level classification (healthy = PAI 1 vs. diseased = PAI 2–5), agreement was excellent ($\kappa = 0.812$, 95% CI $0.698\text{--}0.926$). For ordinal PAI scoring, agreement was almost perfect (weighted $\kappa = 0.918$, 95% CI $0.867\text{--}0.969$). For segmentation masks, the inter-rater Dice similarity coefficient was 0.89 (95% CI $0.84\text{--}0.94$), indicating excellent spatial agreement between annotators and supporting the reliability of the reference standard.

Segmentation performance in periapical lesion detection

Dice similarity coefficients were calculated to evaluate the spatial overlap between predicted and ground-truth lesion boundaries. The training curves demonstrate progressive improvement in segmentation accuracy across all models. DeepLabV3 achieved the highest final Dice score of approximately 0.88, followed by HRNet (0.87), ResUNet34 (0.86), and U-Net (0.84). All models demonstrated stable convergence with minimal overfitting,

Table 2 Prespecified size-based thresholds used to assign aPAI scores from the segmented lesion area ratio

PAI Score	Lesion Area Ratio (% of total pixels)	Description
1	$<0.1\%$	No radiolucent area detected or negligible segmented area
2	$0.1\text{--}0.5\%$	Small segmented radiolucency
3	$0.5\text{--}3\%$	Moderate segmented radiolucency
4	$3\text{--}7\%$	Large segmented radiolucency
5	$>7\%$	Extensive/aggressive segmented radiolucency

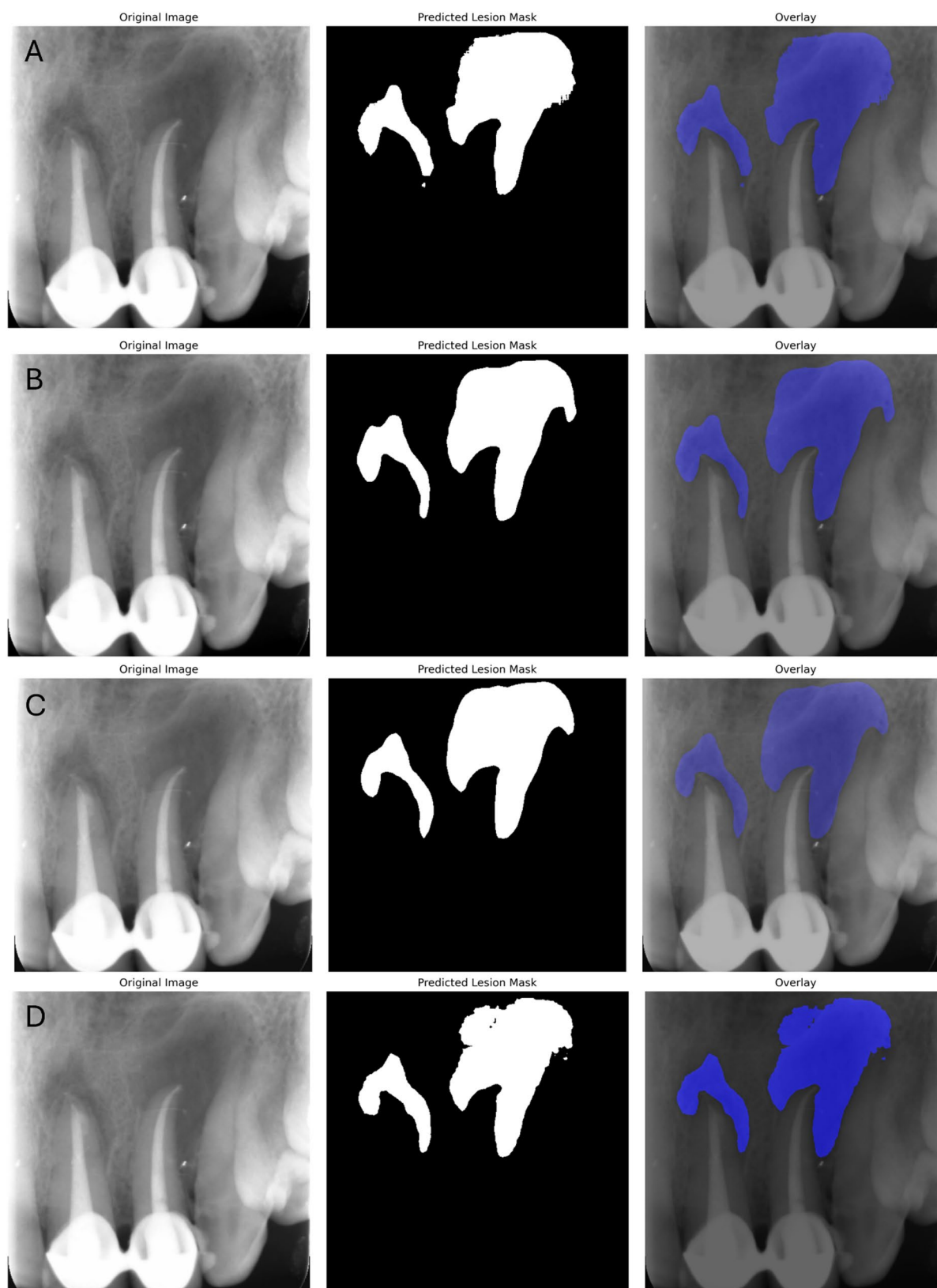


Fig. 2 Representative outputs from the AI models, including U-Net (A), ResUNet34 (B), DeepLabV3 (C), and HRNet (D), showing the original tooth-level input crops, predicted lesion segmentation masks, and overlay images. Details of tooth-level input preparation and manual field-of-view standardization are provided in the Supplementary Methods under “Tooth-level field-of-view definition and manual cropping”

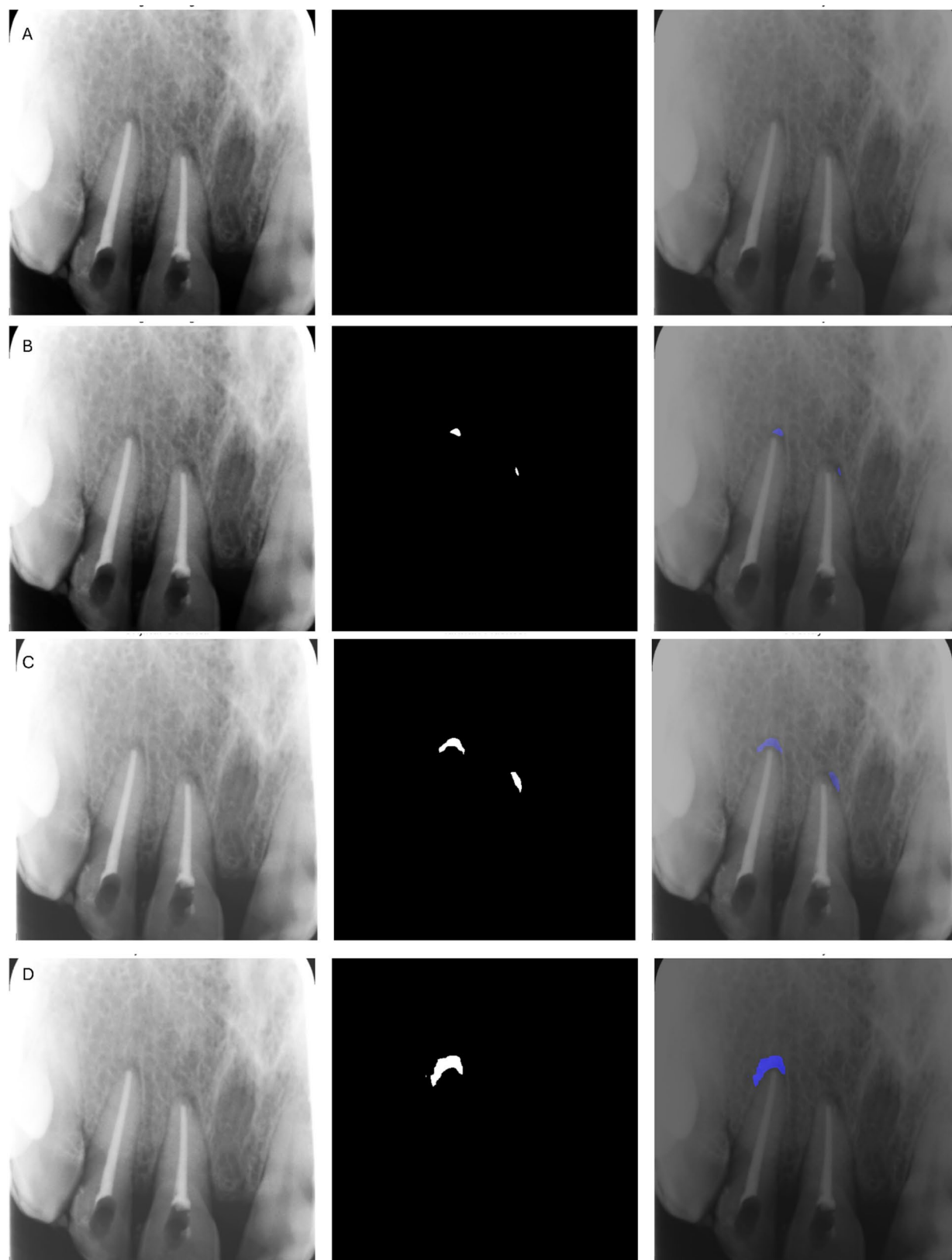


Fig. 3 Representative qualitative examples. Left: original periapical radiograph. Middle: predicted binary lesion mask. Right: overlay of the prediction (blue). Panels A-D illustrate four cases ranging from subtle/small to more evident lesions, as predicted by U-Net (A), ResUNet34 (B), DeepLabV3 (C), and HRNet (D)

as evidenced by the consistent validation Dice scores throughout training (Fig. 4).

Classification performance

Table 3 provides a comprehensive summary of the accuracy, recall (sensitivity), specificity, precision (positive predictive value), and F1-score metrics calculated for each model. Image-level confusion matrices are shown in Fig. 5. For a more granular assessment, the true positive (TP), false positive (FP), true negative (TN), and false negative (FN) counts were determined for all models (Table 3). The test set comprised 161 images, including 97 diseased cases (PAI 2–5) and 64 healthy cases (PAI 1). Furthermore, the diagnostic capabilities of each approach in identifying periapical lesions were analyzed through receiver operating characteristic (ROC) curves, which graphically depict the relationship between sensitivity and specificity (Fig. 6). The area under the ROC curve (AUC) values were as follows: DeepLabV3 (0.94; 95% CI: 0.90–0.97), HRNet (0.92; 95% CI: 0.88–0.96), ResUNet34 (0.91; 95% CI: 0.86–0.95), and U-Net (0.84; 95% CI: 0.78–0.90). All models demonstrated good to excellent discrimination ($AUC > 0.80$), with DeepLabV3 achieving the highest discriminative ability.

Among the four models, DeepLabV3 achieved the highest accuracy (90.1%; 95% CI: 0.85–0.94). This model also achieved the highest recall (92.8%), indicating that it missed the fewest lesions among all models. The highest specificity was observed in the HRNet model (87.5%), meaning it produced the lowest number of false positive classifications. Moreover, HRNet achieved the highest precision (91.4%; 95% CI: 0.84–0.96), demonstrating the most reliable positive predictions. In terms of overall performance balance, DeepLabV3 had the highest F1-score (91.8%). ResUNet34 demonstrated strong performance with 87.0% accuracy, 90.7% recall, and 81.3% specificity. U-Net showed the lowest performance across all metrics, with 77.0% accuracy, 78.4% recall, and 75.0% specificity.

Based on these metrics, DeepLabV3 demonstrated the most balanced and robust performance in detecting lesions, whereas HRNet offered a clinically reliable alternative due to its low false positive rate and highest specificity.

Statistical comparisons

The Friedman test revealed a statistically significant overall difference among the models ($\chi^2(3)=9.82$, $p=0.020$), and the pairwise comparison results are presented in Table 4. Pairwise comparisons using the Wilcoxon signed-rank test with Bonferroni correction for 6 comparisons (adjusted $\alpha=0.0083$) showed that the comparison between DeepLabV3 and U-Net was statistically significant after Bonferroni adjustment ($p=0.0012$; adjusted $\alpha=0.0083$). Likewise, comparisons between

U-Net and ResUNet34 ($p=0.018$) and between U-Net and HRNet ($p=0.012$) suggested numerical trends without statistical significance after correction. No evidence of a significant difference was observed among the three higher-performing models (ResUNet34 vs. DeepLabV3, $p=0.379$; ResUNet34 vs. HRNet, $p=0.873$; DeepLabV3 vs. HRNet, $p=0.472$).

aPAI classification performance

To validate the correspondence between expert-assigned PAI scores and area-based aPAI classifications, quadratic-weighted Cohen's κ was computed for each model on the independent test set ($n=161$). Agreement ranged from substantial to almost perfect: DeepLabV3 ($\kappa=0.939$, 95% CI: 0.906–0.964) and HRNet ($\kappa=0.941$, 95% CI: 0.909–0.965) achieved the highest concordance, followed by ResUNet34 ($\kappa=0.866$, 95% CI: 0.801–0.920) and U-Net ($\kappa=0.839$, 95% CI: 0.772–0.894). Exact category agreement ranged from 72.7% (U-Net) to 84.5% (DeepLabV3), while agreement within ± 1 category exceeded 87% for all architectures, reaching 96.9% for HRNet. These findings demonstrate that the area-based aPAI index provides clinically acceptable correspondence with expert PAI assignments across all tested architectures.

The mean accuracy values observed for aPAI classification differed among the four models and are shown in Table 3: DeepLabV3 (84.5%), HRNet (82.6%), ResUNet34 (80.1%), and U-Net (72.7%). The Friedman test indicated a statistically significant overall difference among the models ($\chi^2(3)=8.76$, $p=0.033$). Pairwise comparisons across all six model pairs were performed using Wilcoxon signed-rank tests with Bonferroni correction (adjusted $\alpha=0.0083$). The comparison between U-Net and DeepLabV3 was statistically significant after Bonferroni adjustment ($p=0.0012$). Comparisons between U-Net and HRNet ($p=0.030$) and between U-Net and ResUNet34 ($p=0.060$) showed numerical trends without statistical significance after correction. No significant differences were observed among DeepLabV3, HRNet, and ResUNet34.

The class-wise prediction accuracies of each model across different aPAI categories are summarized in Supplementary Table 1. All models showed highest accuracy in identifying aPAI 5 cases (88–100%), followed by aPAI 4 (72.7–86.4%) and aPAI 1 (75–87.5%). Performance was lowest for aPAI 2 (60–72%) and aPAI 3 (64–76%), reflecting the clinical challenge of distinguishing early-stage periapical changes. These findings suggest that the models were more reliable in detecting advanced lesions (PAI 4–5) and healthy teeth (PAI 1), whereas mild or early-stage lesions (PAI 2–3) posed a greater classification challenge.

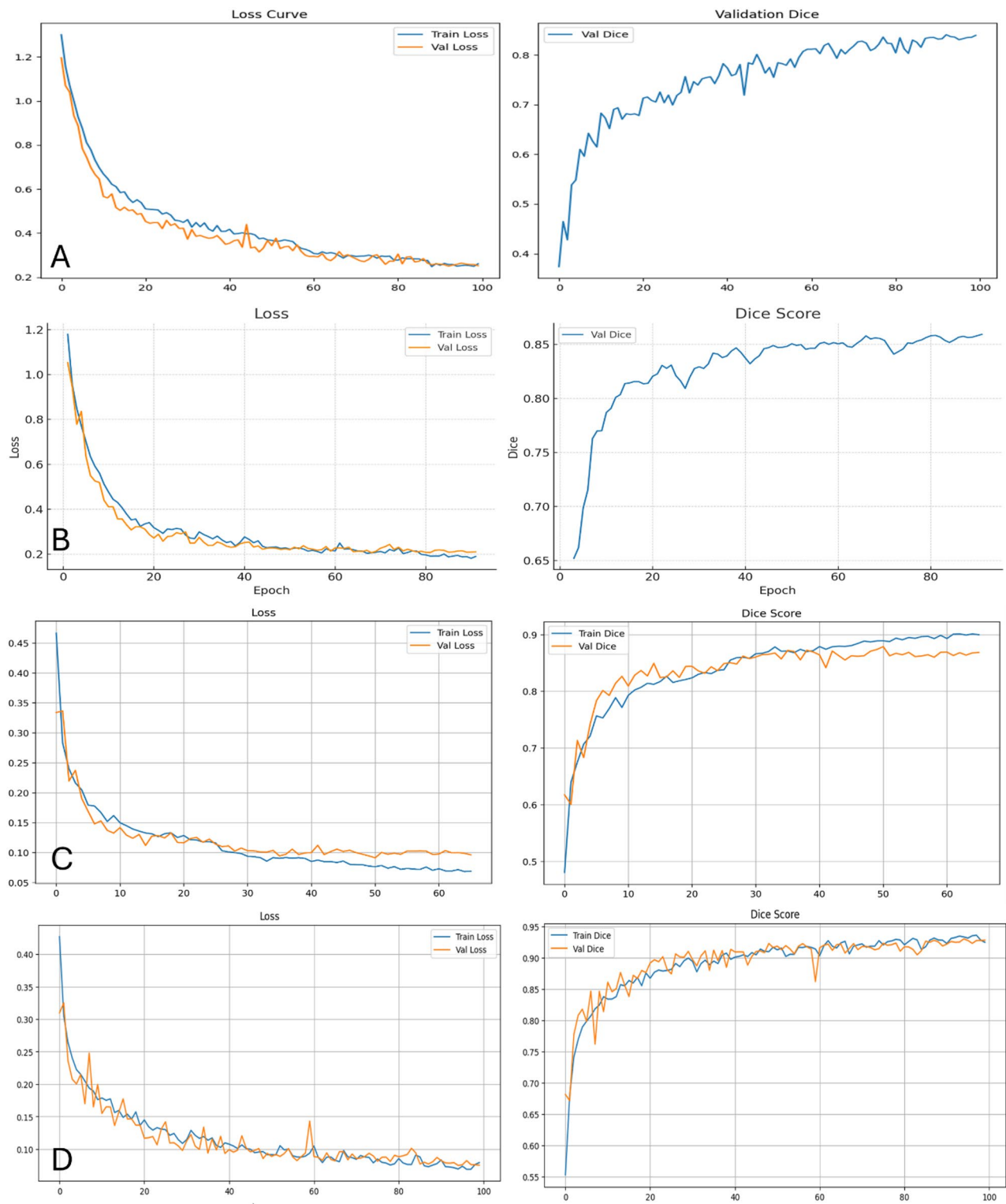


Fig. 4 Comparison of loss and Dice score graphs illustrating the training processes of U-Net (A), ResUNet34 (B), DeepLabV3 (C), and HRNet (D) models

Table 3 Classification metrics for periapical lesion detection by each model on the independent test set ($n = 161$), including accuracy, sensitivity, specificity, precision, F1-score, confusion matrix components (TP, FP, TN, FN), and mean aPAI classification accuracy

Model	Accuracy, % (95% CI)	Sensitivity, % (95% CI)	Specificity, % (95% CI)	Precision, % (95% CI)	F1-score, % (95% CI)	TP	FP	TN	FN	Mean aPAI Ac- curacy, %
U-Net	77.0 (72.8–83.2)	78.4 (70.2–82.6)	75.0 (70.8–81.4)	82.6 (77.4–89.7)	80.4 (76.7–85.3)	76	16	48	21	72.7
ResUNet34	87.0 (82.0–93.6)	90.7 (84.7–93.9)	81.3 (74.3–86.6)	88.0 (83.3–93.8)	89.3 (86.6–92.9)	88	12	52	9	80.1
DeepLabV3	90.1 (85.3–94.8)	92.8 (87.0–95.4)	85.9 (79.7–89.4)	90.9 (86.3–94.7)	91.8 (88.6–94.9)	90	9	55	7	84.5
HRNet	87.6 (81.1–91.1)	87.6 (81.8–91.9)	87.5 (83.2–92.6)	91.4 (88.1–95.2)	89.5 (84.7–93.9)	85	8	56	12	82.6

CI Confidence interval, TP True positive, FP False positive, TN True negative, FN False negative, aPAI Area-based Periapical Index

**Fig. 5** Normalized confusion matrices for image-level lesion detection. Y-axis = True label (Positive = PAI ≥ 2; Negative = PAI = 1). X-axis = Predicted label. Cells show proportions (optionally counts). Operating point: pixel-probability 0.5 with ≥ 131 px (~0.05%) area to call “Positive”. Panels: (A) U-Net, (B) ResUNet34, (C) DeepLabV3, (D) HRNet. Panel C shows higher recall; Panel D higher specificity

Discussion

In this study, we aimed to automatically detect periapical lesions using four DL architectures (U-Net, ResUNet34, DeepLabV3, HRNet) and perform area-based PAI scoring. Based on our comparative analysis, the null hypothesis was rejected, demonstrating statistically significant differences in segmentation accuracy among the architectures for automated periapical lesion detection and PAI scoring.

Early detection of periapical lesions is essential for effective endodontic treatment and maintaining oral health. Apical periodontitis is highly prevalent, with studies indicating that it may affect up to half of the

adult population [10]. Many chronic periapical lesions are asymptomatic, and untreated infections can lead to serious complications [10]. Thus, detecting these lesions on periapical radiographs is essential to prevent disease progression.

Periapical radiographs are the most frequently used modality for detecting periapical pathologies [20]. While CBCT offers higher sensitivity, its routine use is limited by cost and radiation exposure [21]. We selected periapical radiographs as the primary modality because CBCT is not routinely used, while panoramic radiographs have lower resolution and higher risk of anatomical

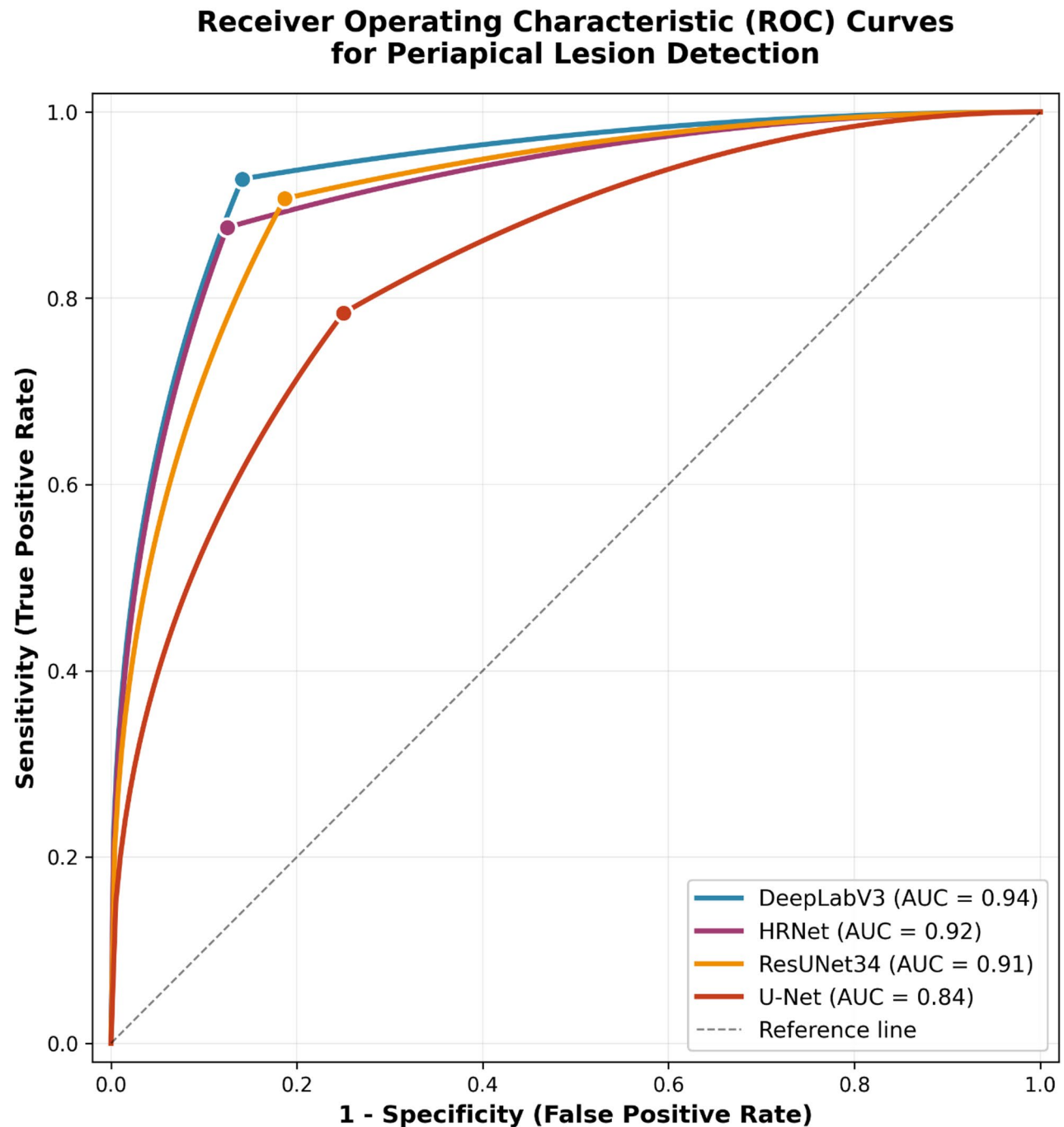


Fig. 6 ROC curves showing diagnostic performance (AUC) of four models for lesion detection

superimposition that may compromise detection accuracy [21, 22].

AI applications have advanced rapidly in medical imaging, with computer-aided diagnosis software enhancing diagnostic speed and accuracy [23, 24]. DL-based CNNs have been the most successful approach. U-Net enables high accuracy medical image segmentation even with limited datasets, while variants such as ResUNet34, DeepLabV3, and HRNet offer distinct advantages:

DeepLabV3 captures multi-scale features through atrous convolutions, while HRNet maintains high-resolution feature maps for fine detail detection. These various CNN architectures enable pixel-level delineation and quantification of periapical lesions, thus providing a more objective assessment compared to traditional approaches. In our study, all four of these DL architectures were evaluated and systematically compared in terms of periapical lesion detection and PAI scoring.

Table 4 Pairwise statistical comparisons between models for lesion detection accuracy using the Wilcoxon signed-rank test. Overall differences among models were assessed using the Friedman test ($\chi^2(3) = 9.82, p = 0.020$; see text)

Comparison	Wilcoxon Z	p-value	Bonferroni ($\alpha = 0.0083$)
U-Net vs. DeepLabV3	-3.23	0.0012	$p < 0.0083$
U-Net vs. HRNet	-2.52	0.012	$p > 0.0083^*$
U-Net vs. ResUNet34	-2.36	0.018	$p > 0.0083^*$
ResUNet34 vs. DeepLabV3	-0.88	0.379	$p > 0.0083$
DeepLabV3 vs. HRNet	-0.72	0.472	$p > 0.0083$
ResUNet34 vs. HRNet	-0.16	0.873	$p > 0.0083$

*Nominal $p < 0.05$ before Bonferroni correction; not statistically significant after correction

Previous studies [7, 25–28] have demonstrated promising AI performance for periapical lesion detection. Bayrakdar et al. [25] reported 92% sensitivity and 88% F1-score using U-Net on panoramic radiographs. Endres et al. [26] showed their model outperformed 14 of 24 oral surgeons, achieving 67% precision and 51% sensitivity. Ketenci Çay et al. [27] achieved ~ 90% accuracy with DeepLabV3 on panoramic images. Liu et al. [28] demonstrated that ConvNeXt outperformed ResNet34 and improved novice dentist performance. A recent study [7] on 2001 tooth segments reported an AUC of 0.85 (sensitivity 0.65, specificity 0.87), which improved to 0.95 for clearly detectable lesions. These findings suggest that the diagnostic performance of DL models can vary according to lesion certainty. In addition, tooth type may influence model sensitivity and overall accuracy. In our study, DeepLabV3 demonstrated superior performance, achieving the highest accuracy (90.1%), sensitivity (92.8%), and F1-score (91.8%). This superiority is attributed to its atrous spatial pyramid pooling (ASPP) architecture, which enables multi-scale feature extraction crucial for detecting periapical lesions of varying sizes. Its dilated convolutions capture both local detail and global context, making it well-suited for dental radiographic analysis. The high sensitivity is clinically significant as it minimizes the risk of missing pathological cases. Dice similarity coefficients validated these findings: DeepLabV3 achieved 0.88, followed by HRNet (0.87), ResUNet34 (0.86), and U-Net (0.84). All models exceeded the 0.8 threshold considered clinically acceptable for medical image segmentation, indicating excellent spatial overlap between AI predictions and expert-annotated boundaries. HRNet achieved the highest specificity (87.5%) and precision (91.4%), producing the fewest false positives and most reliable positive predictions. Both models showed numerically better performance than U-Net. After Bonferroni correction, only the DeepLabV3 versus U-Net comparison remained statistically significant ($p = 0.0012$); differences between U-Net and the other two models showed numerical trends without reaching the

corrected threshold. DeepLabV3 provided the most balanced overall performance, whereas HRNet offered the highest specificity. From a clinical decision-making perspective, DeepLabV3’s high sensitivity (92.8%) reduces the risk of missing treatable periapical pathology, while HRNet’s superior specificity (87.5%) minimizes unnecessary interventions. Model selection should align with clinical priorities—screening applications may favor high sensitivity models, while confirmatory diagnosis may benefit from high specificity architectures.

For aPAI classification in our study, model performance varied notably, with mean accuracy ranging from 73% (U-Net) to 85% (DeepLabV3). The Friedman test revealed statistically significant differences among the models, with U-Net performing significantly worse than DeepLabV3 after Bonferroni correction; differences between U-Net and the remaining two architectures showed numerical trends without reaching the corrected threshold. Class-wise analysis revealed highest accuracy for aPAI 5 cases (88–100%) and aPAI 1 cases (75–88%), reflecting distinct radiographic features of advanced lesions and normal periapical structures. However, performance was notably lower for aPAI 2 (60–72%) and aPAI 3 (64–76%), where subtle trabecular changes posed greater classification challenges. This pattern likely reflects both the inherent difficulty of early lesion detection and the fundamental limitation of area-based scoring in capturing qualitative features that distinguish PAI 1 from PAI 2.

The 5-category PAI system developed by Ørstavik et al. [13] served as the conceptual framework for our automated scoring approach. However, it is essential to distinguish between the traditional PAI system and our aPAI. Traditional PAI assessment relies on qualitative radiographic features—lesion boundary characteristics (well-defined vs. diffuse margins), internal radiolucency patterns (homogeneous vs. heterogeneous), trabecular bone architecture changes, and surrounding bone density alterations and is subject to interobserver variability and subjectivity. In contrast, our aPAI derives scores solely from lesion-to-image area ratios, providing quantitative, reproducible measurements but cannot capture these qualitative visual assessments central to traditional PAI scoring. While automated aPAI scoring offers the potential for more consistent evaluation through objective area measurements, several inherent limitations must be acknowledged. First, the classification thresholds (Table 2) were prespecified based on theoretical relationships between PAI categories and expected lesion dimensions, rather than derived from training data, to avoid information leakage. While this approach ensures methodological rigor, the thresholds represent approximations that may not optimally correspond to expert PAI assignments in all cases. Second, using the total image area as the

denominator introduces composition-dependent variability; images with different cropping, angulation, or field-of-view settings may yield different aPAI scores for identical lesions. In addition, the lack of pixel-to-millimeter calibration means that aPAI scores represent relative rather than absolute lesion dimensions. While this is acceptable for within-study comparisons under consistent acquisition and preprocessing protocols, external validation studies should incorporate pixel spacing metadata and/or phantom-based calibration to ensure physical measurement consistency across different imaging systems and institutions. Third, qualitative features often carry diagnostic significance independent of lesion size, particularly for distinguishing early periapical changes (PAI 1–2 boundary) where subtle trabecular disruption may precede measurable radiolucent areas. The reduced performance in mild categories observed in our study was consistent across all four architectures, suggesting that these limitations reflect inherent challenges in area-based quantification of subtle periapical changes rather than deficiencies specific to any modeling approach. Previous studies [29, 30] support these patterns: Moidu et al. [29] demonstrated that a CNN-based model achieved 86.3% overall accuracy for PAI scoring on periapical radiographs, with best performance for healthy cases but lower prediction rates for borderline categories. Similarly, Arijji et al. [30] reported high sensitivity (0.88) for detecting mandibular radiolucent lesions, though detection was less reliable for small lesions with poorly defined borders—a pattern consistent with the inherent difficulty of quantifying subtle periapical changes.

From a clinical workflow perspective, aPAI should be considered a radiographic decision-support and severity-stratification tool rather than an autonomous treatment-planning system. In practical terms, aPAI 1 may support a radiographic impression of normal periapical status when consistent with the absence of clinical signs and symptoms. Intermediate categories (aPAI 2–3) should be interpreted cautiously, as they may reflect subtle or borderline radiographic change and therefore warrant correlation with symptoms, pulp sensibility findings, treatment history, and follow-up imaging rather than immediate treatment decisions based on radiography alone. Higher categories (aPAI 4–5), which were also the most reliably identified in our study, may support classification of more established periapical disease, facilitate standardized case documentation, and help prioritize closer diagnostic evaluation, retreatment assessment, or advanced imaging when conventional radiographic findings remain uncertain. Accordingly, the principal clinical value of aPAI lies in improving consistency in radiographic grading, longitudinal comparison, and communication across observers, while final diagnosis and

treatment planning must remain based on comprehensive clinical and radiographic interpretation.

A fundamental consideration in pipeline-based approaches is that downstream aPAI classification inherently depends on upstream segmentation quality. As demonstrated in recent work on automated dental structure quantification [31], suboptimal segmentation can propagate errors to subsequent measurements. Analysis of all discordant cases where model-derived aPAI differed from expert PAI by at least one category ($n = 44$, 27.3% of test set) revealed that segmentation-related factors accounted for all misclassifications. Specifically, area quantification errors (over- or under-estimation of lesion boundaries) were responsible for 84.1% of discordant cases, with area over-estimation being the predominant pattern (63.6%), while binary detection failures (missed or falsely detected lesions) contributed to the remaining 15.9%. These findings indicate that segmentation boundary precision—not merely binary detection accuracy—is the primary determinant of aPAI reliability and underscore that achieving high segmentation quality is a prerequisite for dependable area-based quantification.

Future methodological refinements could enhance clinical validity. Defining a standardized apical region of interest (ROI) as the reference area would reduce cropping variability. Specifically, a tooth-centric denominator could be implemented by automatic apex localization using a dedicated CNN, defining a fixed-size ROI centered on the detected apex, and computing the lesion area ratio within this standardized ROI. Such normalization would substantially reduce dependence on variable image framing and improve reproducibility across different cropping protocols and imaging systems; however, it would require additional model development and validation for robust apex detection, which we propose as a priority for future work. Utilizing continuous probability maps (0–1 range) instead of binary masks could better capture boundary uncertainty and subtle changes. Finally, calibrating aPAI thresholds against CBCT volumetric measurements in approximately 60–100 paired cases would strengthen correspondence with established staging systems.

Recent AI studies [7, 8, 25, 27, 28, 30, 32–34] on dental radiographs have mainly addressed lesion detection, localization, or image-level classification. In contrast, the present study extends that literature in three critical ways: by implementing a fully automatic, tooth-level segmentation workflow on periapical radiographs; by systematically comparing four different DL architectures under identical preprocessing and evaluation conditions—including models like HRNet and ResUNet34 that have rarely been explored in dental radiology; and by translating these segmentation outputs into a rule-based, area-derived PAI proxy. This design allowed us to examine not

only whether periapical lesions can be detected, but also whether lesion extent can be converted into a clinically interpretable and reproducible grading framework. In that respect, the novelty of our study lies less in applying AI to lesion detection alone, and more in proposing and comparatively validating a structured, objective bridge between AI-based segmentation and clinically relevant PAI-like radiographic grading.

Several limitations should be considered when interpreting the present findings. First, as a retrospective single-center study utilizing archived radiographs acquired under standardized institutional conditions (KaVo Focus digital intraoral X-ray unit, 70 kVp, 7 mA; Digora PSP plates), the potential for spectrum bias exists. Consequently, while our results establish a strong baseline, the model's performance requires further validation in populations with diverse lesion prevalences, referral patterns, and imaging ecosystems to fully confirm its external generalizability. Although the paralleling technique was employed to minimize geometric distortion, slight variations in projection geometry across images may still influence apparent lesion dimensions. In addition, because aPAI normalizes lesion size by the total image area, it is sensitive to field-of-view and cropping variability; despite standardized cropping, residual anatomical differences across cases could introduce systematic variations in aPAI estimates. Exploring alternative normalization strategies, such as using a fixed periapical region-of-interest, may help mitigate this dependency in future applications. Second, because predictive values (PPV and NPV) are prevalence-dependent, the metrics observed in our 60% diseased cohort should be interpreted within that context rather than directly extrapolated to lower-prevalence screening settings. Third, grouping all periapical radiolucencies without differentiating by specific lesion types or tooth categories may influence performance in certain clinical scenarios, and real-time deployment would necessitate further integration with automatic tooth localization systems. Finally, 2 methodological considerations regarding the reference data should be acknowledged. First, because the generation of robust reference masks required lesions with clearly discernible boundaries, very subtle or borderline early lesions may have been less frequently represented in the annotated dataset. Although this approach was necessary to ensure annotation consistency and mask reliability, it may have favored cases with greater radiographic conspicuity and, therefore, slightly enhanced segmentation agreement relative to fully unselected clinical conditions. Second, the reference standard was established through expert consensus. This strategy strengthened annotation quality and methodological rigor; however, as with all expert-derived reference standards in medical imaging research, a degree of observer-dependent variability cannot be entirely

excluded. In this respect, future multicenter investigations incorporating complementary objective volumetric assessments from CBCT may further refine and extend the framework presented in the current study.

From a clinical perspective, the proposed workflow should be regarded as a decision-support framework rather than an autonomous diagnostic tool. By translating segmentation outputs into standardized PAI-like categories, it may improve consistency in radiographic grading, support longitudinal follow-up, and facilitate communication across observers. DeepLabV3 appears particularly suitable for screening-oriented scenarios because of its high sensitivity, whereas HRNet may offer advantages when minimizing false-positive findings is prioritized. Overall, the main contribution of this study lies in establishing a clinically interpretable bridge between AI-based lesion segmentation and reproducible radiographic severity stratification.

Taken together, these findings suggest that the principal contribution of the present study is not to replace conventional PAI scoring, but to provide a standardized segmentation-based framework that translates radiographic lesion extent into a reproducible PAI-like severity scale. In this respect, the workflow may be most valuable for structured follow-up assessment, interobserver consistency, and research standardization, while clinical decision-making should remain based on integrated radiographic and clinical interpretation.

Conclusions

In conclusion, DL-based periapical lesion segmentation combined with area-based aPAI scoring shows promise as a clinically supportive tool for radiographic assessment. Rather than replacing clinician judgment or established diagnostic frameworks, this approach may help standardize lesion detection, grading, and follow-up comparisons on routine periapical radiographs. DeepLabV3 appears advantageous for sensitivity-oriented screening scenarios, whereas HRNet may be preferable when minimizing false-positive findings is prioritized. Further multicenter external validation and clinically integrated workflow studies are needed before routine implementation.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12903-026-08645-4>.

Supplementary Material file 1. Additional information is provided in the online supplementary material. Supplementary Methods contain an expanded description of the dataset selection, reference standard annotation protocol, image preprocessing, model architectures, training strategy, and evaluation procedures. Supplementary Table 1 summarizes the main characteristics of the study dataset and detailed performance metrics for each model. A compressed archive (codes.rar) includes the scripts used for data preprocessing, model training, and statistical analysis, while

another archive (data sample.rar) provides an anonymized sample of the radiographs and corresponding masks used in this study. The completed STARD-AI checklist documenting adherence to reporting guidelines for AI-based diagnostic studies is also provided as a separate supplementary file

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Authors' contributions

YEC: Conceptualization; Methodology; Software; Validation and formal analysis; Investigation and data curation; Writing – original draft; Writing – review and editing; Visualization; Resources.DE: Conceptualization; Methodology; Validation and formal analysis; Investigation and data curation; Writing – original draft; Writing – review and editing; Resources.KE: Conceptualization; Writing – review and editing; Supervision; Resources.AK: Writing – review and editing; Supervision; Project administration; Resources.All authors read and approved the final manuscript.

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Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to institutional and ethical restrictions; anonymized training images are included in the supplementary file. Anonymized sample images, model training configurations, and analysis scripts supporting the findings of this study are also available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This retrospective study was conducted in accordance with the Declaration of Helsinki and was approved by the Clinical Research Ethics Committee of Akdeniz University Faculty of Medicine (Approval No. TBAEK-624, 03 July 2025). In accordance with institutional policy and applicable regulations, written informed consent was obtained from all patients at the time of their presentation to the faculty, authorizing the use of their anonymized clinical and imaging data for scientific research and educational activities.

Consent for publication

This manuscript does not contain any individual person's identifiable data in any form (including images or clinical details). Consent for publication of anonymized radiographic images was obtained as part of the general institutional consent procedure.

Competing interests

The authors declare no competing interests.

Declaration of Generative AI

The authors used OpenAI ChatGPT solely as a basic AI-assisted spelling and grammar checker to identify potential language issues. No generative AI tools were used for writing, content creation, data analysis, or drawing scientific conclusions. The authors take full responsibility for the content of this manuscript.

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